

albattikha@ucsd.edu
(619) 897-4800

Alex Battikha

linkedin.com/in/alex-battikha/
US Citizen

EDUCATION

University of California, San Diego

B.S. in Bioengineering

San Diego, CA

Expected June 2029

- **Jacobs Scholar:** A prestigious merit scholarship awarded to sub-1% of engineering applicants.
- **Concentrations:** Biosystems & Economics
- **Coursework:** Control Systems, Computer Vision, Data Structures & Algorithms, Object-Oriented Programming, Circuits & Electronics, Linear Algebra, Multi-Variable Calculus, Physics I & II (Mechanics, E&M)

WORK EXPERIENCE

Sanford Burnham Prebys Institute

Research Intern

San Diego, CA

June 2024 – Present

- Co-lead developer on ecPATH, a novel pipeline detecting extrachromosomal DNA from histopathology images.
 - Collaborating with postdoctoral researchers to improve accuracy, validate results, and address revisions for pending *Nature* publication.
- Programmed 4 machine learning models with Python and optimized pipeline performance and reproducibility through model tuning, preprocessing improvements, and workflow standardization for clinical adoption.
- Elicited feedback from oncologists to optimize the model in preparation for providing nearby hospitals with access.

Johns Hopkins University Research Institute

Research Intern

Baltimore, MD

May 2024 – May 2025

- Worked alongside undergraduate students and Dr. Dahbura at the Johns Hopkins University Information Security Institute to develop a novel, *patent-pending* robotics communication protocol.
- Engineered an autonomous robotic testbed to validate a patent-pending protocol for wireless distribution systems by programming the full software stack, from low-level path-following to high-level task logic, in Python.
- Compiled results and presented at Johns Hopkins University Design Day.

FIRST Tech Challenge (FTC)

Team Captain

San Diego, CA

Oct 2019 – April 2025

- Managed a team of 14 high school students in developing a full-scale robot during a 25-week time period in preparation for the FTC competition.
- Led the mechanical sub-team in the design, fabrication, and assembly of the various hardware components of the robot, resulting in a mechanical performance reliability rate over 99.1%.
- Led community outreach, team building, and mentorship events with companies like NASA, Google, and Amazon.

PROJECTS

- **Autonomous UAV for Environmental Remediation:** Developed a full-scale, unmanned aerial vehicle (UAV) system to autonomously detect oceanic pollution, specifically addressing challenges in scalable software architectures. Achieved 96% precision on a 6000+ image dataset by implementing and optimizing a YOLOv8 vision detection pipeline.
- **Cryptographic PUF Robustness Analysis:** Under the mentorship of UCLA Professor Potkonjak, developed a novel physically unclonable function (PUF) to enhance cryptographic security. Validated system robustness and decryption resistance using Python, NIST Statistical Tests, and AI-driven performance analysis.

INVOLVEMENT

- **UCSD Advanced Robotics & Controls Lab:** Contributed to LapSurgie, a novel framework for performing teleoperated laparoscopic surgery using a humanoid robot. The work involves optimizing the remote center of motion (RCM) by integrating a time-of-flight sensor and visual laser to minimize tool-tip error and enhance surgical precision.

HONORS & AWARDS

- **T25 Teen:** Named San Diego County's 25 Most Remarkable Teens.
- **FIRST Tech Challenge (1st Place):** World Championship Winner (1st place out of 7100+ teams).
- **U.S. Youth Delegate:** Selected as 1 of only 9 U.S. youth delegates for a UN-affiliated environmental conference.

SKILLS

Languages: Python, Java, Javascript, C++, C, MATLAB

Technologies: Git, SolidWorks, Solidworks Simulation, ROS, OpenCV, TensorFlow, RaspberryPi, Linux (Ubuntu)